NOTE

(i) λ is called the parameter of the Poisson distribution.

(ii) $e = 1 + \frac{1}{1!} + \frac{1}{2!} + ... = 2.71828\ldots$ is an irrational number.[n]

Characteristics of Poisson Distribution

(i) Poisson distribution is a discrete distribution i.e., X can take values 0, 1, 2,…

(ii) p is small, q is large and n is indefinitely large i.e., $p \longrightarrow 0$ $q \longrightarrow 1$ and $n \longrightarrow \infty$ and np is finite

(iii) Values of constants : (a) Mean = λ = variance (b) Standard deviation = $\sqrt{\lambda}$ (c) Skewness = $\frac{1}{\sqrt{\lambda}}$ (iv) Kurtosis = $\frac{1}{\lambda}$

(iv) It may have one or two modes

(v) If X and Y are two independent Poisson variates, $X+Y$ is also a Poisson variate.

(vi) If X and Y are two independent Poisson variates, $X-Y$ need not be a Poisson variate.

(vii) Poisson distribution is positively skewed.

(viii) It is leptokurtic.

NOTE

As p is very small the probability of happening of the event is rare and this distribution is applicable where study of interest is on rare events

Some examples:

(i) The event of a student getting first mark in all subjects and at all the examinations

(ii) The event of finding a defective item from the production of a reputed company

(iii) The number of blinds born in a particular year

(iv) The number of mistakes committed in a typed page

(v) The number of traffic accidents per day at a busy junction.

(vi) The number of death claims received per day by an insurance company.